LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Installation and operation of Polarized relay.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Installation and operation of Polarized relay.
5. Specific Objectives	24.1.1.1 Trainee will be able to explain the operation of Polarized relay and contacts.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of relay coil energization and contact closure; Safety: Ensure power is off before touching terminals; Instructor supervision: Monitor trainee observation of polarity effects.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a relay?
(b) How does a polarized relay differ from a standard relay?
(c) Why is polarity critical in polarized relay operation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Construction of polarized relay • Permanent magnet assembly • Soft iron armature details • Contact arrangement types • Polarity sensitivity mechanism	L & Ds	PPT, IB	15 min
(b)	• Operation on positive polarity • Operation on negative polarity • Neutral position behavior • Contact movement sequence • Magnetic flux direction effects	L & Ds	PPT, PDF	20 min
(c)	• Installation mounting requirements • Wiring polarity verification • Terminal identification marks • Secure connection techniques • Final operational check steps	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how reversing the coil current affects the armature position.
(b) Describe the steps required to verify correct polarity during installation.
(c) What happens to the contacts if the relay is operated without defined polarity?

12. Home Assignment and Follow Up	Prepare a brief note comparing the installation requirements of a polarized relay versus a standard contactor to be discussed in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Installation and operation of contactor.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min